# An Uncertainty-Aware Smart Irrigation Framework using SAR-Enhanced Soil Moisture Prediction and Batch Scheduling


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***Abstract*—Irrigation scheduling is an essential requirement for sustainable agriculture, especially in arid regions with limited rainfall and heterogeneous water requirements for different crops. The conventional irrigation systems are based on thresholds that may not incorporate factors such as forecasted weather, satellite remote sensing, uncertainties, feasibility, and operational simplicity. In this paper, a smart irrigation scheduling algorithm is proposed that considers multiple inputs including Sentinel-1 Synthetic Aperture Radar (SAR) observations, machine learning forecasts of future soil moisture levels, uncertainty in decision making, weather forecasts, adjustment of crop coefficients, and batch scheduling. Sentinel-1 VV and VH backscatter data are extracted from GeoTIFF images, converted to soil moisture levels, and corrected using a weighted approach in addition to the soil moisture values in datasets. Multiple machine learning algorithms including Random Forests, Gradient Boosting Trees, and Long Short-Term Memory (LSTM) are tested in order to find the best predictor of future soil moisture level using current rainfall and temperature levels as well as lagged variables and SAR-corrected soil moisture. In this model, uncertainty is captured based on the variance between different trees within an ensemble model, which is used in the decision making process to prevent under-irrigation. Results indicate that the Smart SAR + Uncertainty scheduler uses less water than the basic rule-based system without any soil moisture stress days. In addition, using batch scheduling leads to lower irrigation frequencies.**



***Index Terms*—Smart Irrigation, Soil Moisture Prediction, Synthetic Aperture Radar, Sentinel-1, Random Forest, Gradient Boosting, LSTM, Uncertainty-Aware Scheduling, Precision Agriculture, Crop Coefficient**


## I. INTRODUCTION

Irrigation in agriculture consumes large amounts of fresh water sources, accounting for about 70% of global freshwater use [1]. Irrigation inefficiency causes serious losses due to water wastage and stress of the plants. The majority of traditional irrigation techniques use pre-set routines, human judgement, or simplistic controllers based on simple sensors measuring soil moisture level [12]. Despite being relatively easy to employ, such irrigation systems are purely reactive and do not take into account weather forecast, crop-dependent watering needs, and uncertainties related to future state of the soil moisture. As a consequence, farmers tend to over-irrigate their fields resulting in excess water consumption, leaching of nutrients, and saturation of the roots, or under-irrigate, which leads to plant stress [10]. Remote sensing data and advanced machine learning algorithms give numerous advantages in terms of improving farm irrigation system. Firstly, satellite imagery such as SAR data can be used for estimating soil moisture content because microwave backscatter depends on soil dielectric properties and surface wetness, as well as presence of plant coverage [5]. In contrast to other optical sensors like Sentinel-2 or Landsat imagery, SAR is available even at night and during cloudy conditions making it an excellent tool for monitoring agricultural fields [2].However, backscattering measured from SAR images cannot be considered volumetric soil moisture without further calibration [6]. At the same time, machine learning methods have shown impressive results in modeling complicated and non-linear temporal interdependencies between the meteorological factors (e.g., precipitation and temperature), past soil moisture lags, and future changes in moisture [11].Although traditional regression-based methodologies like Support Vector Regression (SVR) and Random Forest (RF) [4] were always preferred due to high reliability, there are some modern developments in machine learning techniques, like Deep Learning based LSTM [8] or GBM (Gradient Boosting Machine) [7]. The choice of the best tradeoff between accuracy and computation speed required by IoT edge-computing solutions is an important consideration that must be taken into account.

The existing research in the area often tends to focus on separate issues concerning the precision agriculture, namely soil moisture measurement, satellite-based retrieval, or IoT irrigation control. However, only a few studies cover the complete process of scheduling from remote sensing to multi-model prediction, uncertainty-driven decision-making, phenotypic adaptability, and, finally, practical scheduling procedure. This study seeks to fill in the gap in the literature and proposes an IoT-based approach for soil moisture monitoring and smart irrigation, which includes comparison between several machine learning algorithms and their applications for scheduling purposes.

The major contributions of the work have been listed below:

- A comprehensive SAR processing pipeline that extracts VV and VH backscatter from Sentinel-1 GeoTIFF imagery and calibrates a SAR-derived proxy into volumetric soil moisture estimates using an empirical linear calibration approach.
- An optimized weighting approach that combines SAR-derived moisture readings with ground-based soil moisture data sources in order to generate a more reliable moisture measurement independent of instrumentation type.
- A detailed analysis of three different deep learning approaches (Random Forest, Gradient Boosting, and LSTM Networks) for predicting soil moisture over multiple days based on live API weather data, as well as incorporating temporal lagged data.
- An irrigation scheduler with built-in uncertainty quantification that utilizes the variance at each decision tree level in ensemble models in order to generate a conservative moisture prediction, favoring the survival of crops.
- An irrigation optimization layer that batches machine-learning-based recommendations to minimize pump activation frequency, thereby reducing unnecessary mechanical activity.
- A crop-based method that incorporates FAO-56 ($K_c$) crop coefficients and plant phenological stages in order to further refine soil moisture recommendations.

The rest of this paper proceeds as follows. In section II, we provide a review of the existing literature with respect to smart irrigation, SAR-based soil moisture prediction and machine learning-based forecast approaches. In section III, we present our methodology with emphasis on the machine learning mathematical model and the system architecture. In section IV, we discuss the experiments conducted with details of the hyperparameters used and the methodology followed. Section V contains the detailed discussion of the obtained results.

## II. Related Work

Over the past ten years, intelligent irrigation systems have received much attention, owing to the development of IoT-based soil moisture sensors, accessible weather forecast data on cloud-based services, satellite imaging technology, and machine learning algorithms. [12].

### *A. IoT and Edge Computing in Agriculture*

Sensor-driven systems provide direct and high-resolution information about the soil profile [11], [12]. The study carried out by Gutiérrez et al. [12] showed that automated irrigation using wireless sensor network (WSN) was an efficient approach. However, the implementation of such solutions is very expensive owing to the need for constant recalibration, and these sensors are often incapable of capturing the vast spatial heterogeneity of soils in large areas of agricultural land [11]. Although edge computing technologies have strived to eliminate reliance on cloud solutions through local computation, they too have been restricted by the narrow spatial reach of physical sensors [11].

### *B. SAR-Based Soil Moisture Retrieval*

The remote sensing method makes up for the deficiencies of the in-situ method as far as the spatial scale is concerned, as remote sensing allows macro-level observations [5]. Sentinel-1 SAR satellite data, which is managed by the European Space Agency [2], has become popular due to its sensitivity to dielectric properties of soil. SAR-based methods for the estimation of soil moisture usually involve sophisticated data processing and calibration procedures to correct for surface roughness and vegetation scattering [5]. The use of the Vertical-Vertical (VV) polarization mode is justified by its strong correlation with soil moisture content, while the Vertical-Horizontal (VH) polarization allows modeling volume scattering caused by vegetation. The Water Cloud Model (WCM) developed by Attema and Ulaby [6] has become a classical semi-empirical technique for isolating vegetation impact on the backscattered signal. Many current researches utilize various machine learning models along with SAR-derived physical parameters to avoid using computationally expensive radiative transfer models and greatly improve results [5]. Chaudhary et al. [14] explored several machine learning models specifically for soil moisture estimation using Sentinel-1 SAR observations. Even though optical instruments like Sentinel-2 can be used to generate accurate vegetation index values (NDVI, for example), their performance can still be significantly reduced due to clouds and atmospheric disturbances, which makes SAR of Sentinel-1 preferable [2], [5].

### C. Machine Learning in Soil Moisture Forecasting

There have been various machine learning approaches employed for hydrology and agriculture forecasting [10], [11]. Random Forest (RF) [4] is commonly used in table-formatted environmental datasets due to its inherent ability to capture non-linearity, resistance to noise from sensors, few hyperparameters to tune, and excellent prediction results even on medium-sized datasets [10]. The Gradient Boosting algorithms, introduced by Chen and Guestrin through their framework known as XGBoost [7], are a different set of tree models that minimize residuals iteratively but usually result in smaller RMSEs at the expense of increased training complexity. Concurrently, deep learning techniques including Long Short-Term Memory (LSTM) [8] and Gated Recurrent Unit (GRU) [9] networks have brought an incredible transformation to time series prediction tasks. Togneri et al. [13] evaluated the efficacy of various machine learning models for forecasting soil moisture for smart irrigation systems. The LSTM architectures have gates that enable capturing long-term dependencies within the time series related to meteorological conditions, for instance, prolonged periods of drought and monsoon effects. Nevertheless, these architectures demand more training data sets, rigorous tuning of hyperparameters, and increased computational capability at inference time. This research project involves an exhaustive comparison among RF, GBM, and LSTM models to establish their architectural merits.

### D. Risk-Aware and Practical Irrigation Scheduling

The majority of recent research studies on irrigation scheduling suggest using one of two types of irrigation scheduling strategies – a strategy based on rules or a strategy based on deterministic predictions [1], [11], [12]. The former is more explainable, but it is also purely reactive because an irrigation event occurs only when the soil moisture becomes lower than some threshold value. A strategy based on deterministic predictions is more advanced in terms of being able to predict the soil moisture depletion by taking into account upcoming weather factors and past trends [4], [7], [8]. Nonetheless, many of the available approaches are deterministic and depend solely on machine learning predictions without considering their uncertainty. To overcome this constraint, a novel uncertain irrigation scheduler, where variance in the outputs of the ensemble trees serves as an indicator for the level of prediction uncertainty, has been developed. Given that each random forest model produces several independent outputs from the trees generated within, this spread can be seen as an indicator of prediction certainty [4]. In place of using the mean predicted moisture in the soil as the sole input in determining how to irrigate the crops, a risk-based correction mechanism has been introduced to account for the uncertainty involved. In addition, current smart irrigation systems are more concerned about the accuracy of predictions rather than taking into consideration the operational challenges faced due to irrigation infrastructure in real life situations [12]. Small irrigation suggestions despite being mathematically correct may lead to frequent switching of pumps as well as unrealistic irrigation schedules. In order to fill in this gap between theoretical algorithms for scheduling irrigation and the practical applications on the farms, this work has proposed an additional component in the form of a batch irrigation scheduling approach.

### E. Soil Moisture Representations: VWC vs. SMP

In agricultural physics and hydrology, soil water status is primarily quantified using two different metrics: Volumetric Water Content (VWC) and Soil Matric Potential (SMP) [10], [13]. VWC measures the volume of water relative to the total soil volume, representing the quantitative water volume in the crop root zone. In contrast, SMP represents the suction force or tension required by plant roots to extract water, usually measured in kilopascals (kPa). Although crop physiological limits (such as field capacity and permanent wilting point) are physically determined by SMP, conversion between VWC and SMP requires empirical soil water retention curves, which introduce high non-linear parameter errors and soil-specific uncertainties [10], [13]. Therefore, direct modeling of soil moisture in VWC is preferred to avoid error propagation.

### F. Contextual Anomaly Detection and Feature Engineering

Data-driven agricultural systems frequently suffer from anomalous or missing data due to sensor faults, maintenance events, or rare meteorological anomalies. To address this, unsupervised architectures like Autoencoders are used to capture the normal daily behavior of environmental parameters [13]. By calculating the reconstruction loss of an asymmetric autoencoder, an Abnormal Context Index (ACI) is derived. The ACI serves as an indicator of atypical but true context, helping predictive models distinguish between sensor anomalies and genuine hydrological changes, thereby improving the robustness of the forecasting pipeline.

## III. METHODOLOGY

### A. Overall System Architecture

The architecture for this system is envisioned as a modular, end-to-end decision-making pipeline for precise irrigation. This includes data ingestion, mathematical SAR processing, machine-learning-based soil moisture estimation, risk-based decision making, phenotypic modification of crops, and scheduling using machinery. The overall architecture of this system is shown in Fig. 1. The system works non-stop. The historical data for soil moisture, precipitation, temperature [3], SAR signals for Sentinel-1 satellites [2], and agricultural parameters [1] are collected and temporally synchronized. The SAR imagery is processed in order to derive VV and VH matrices which are further corrected based on empirical calibration [6] to derive volumetric soil moisture values. These are delayed in time and supplied to the Machine Learning module along with the forecast provided by the Open-Meteo API system [3]. After that, the output data is passed to the uncertainty-aware scheduler and scaled based on FAO crop coefficients using Batch Scheduler.

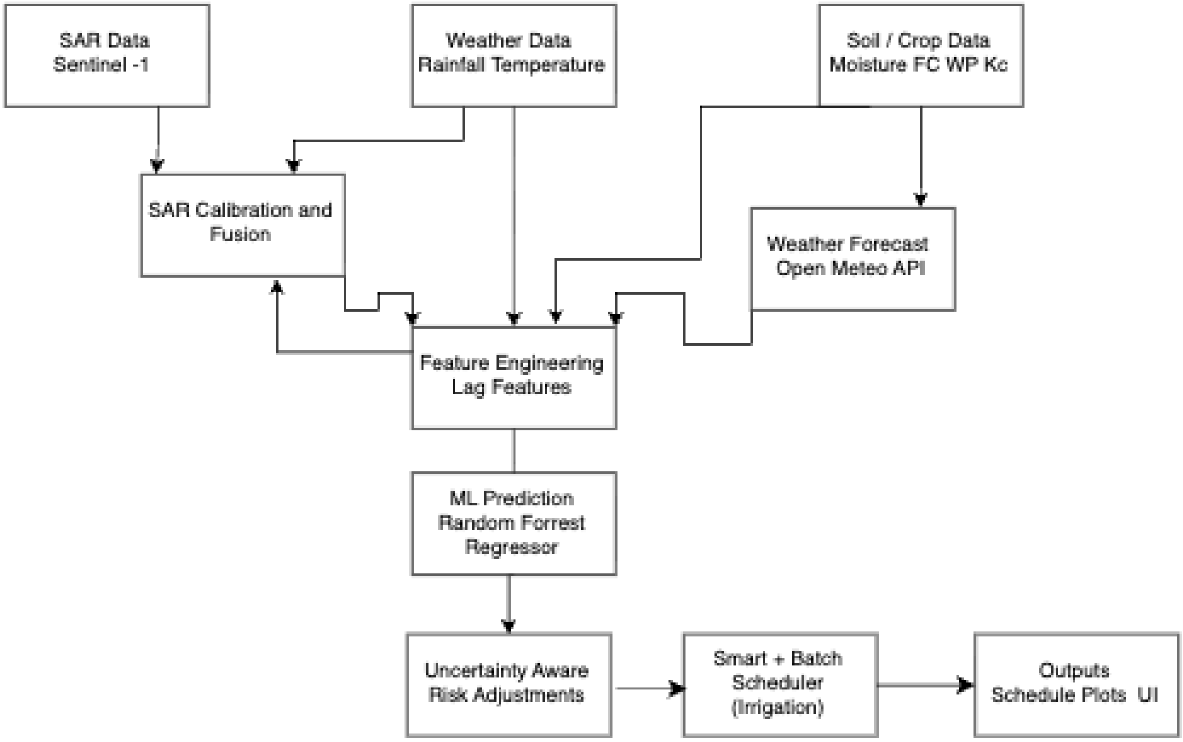


Fig. 1. Overall architecture of the proposed smart irrigation system integrating weather data, Sentinel-1 SAR, soil/crop data, machine learning prediction, uncertainty-aware scheduling, and output generation.

*B. Data Collection, Preprocessing, and Lag Engineering*

The framework relies on three primary categories of high-fidelity data:

1) **Meteorological Data:** Daily rainfall (mm) and temperature (°C) records. Live 5-day forecasts are dynamically requested via the Open-Meteo REST API [3] using geographic coordinates.
2) **Pedological Data:** Ground-truth dataset-based soil moisture values alongside agronomic soil properties such as Field Capacity (FC), Permanent Wilting Point (WP), and Total Available Water (AW) [1].
3) **Remote Sensing Data:** Processed Sentinel-1 SAR GeoTIFF files containing calibrated Ground Range Detected (GRD) VV and VH backscatter coefficients (measured in decibels, dB) [2].

Data preprocessing consists of aligning temporally and interpolating missing values using linear splines. For the machine learning algorithms to recognize the inertia inherent in soil hydrology, lag variables are created. For each variable $X \in \{Rainfall, Temperature, SoilMoisture\}$, we construct a sliding historical window:

$$X_{lag\_k}(t) = X(t-k), \quad \forall k \in \{1, 2, ..., 7\} \tag{1}$$

This 7-day memory provides the ML models with critical context regarding recent drying trends and cumulative precipitation. The complete feature engineering pipeline, detailing the conversion of raw meteorological and soil moisture datasets into temporal lags, is shown in Fig. 2.

*C. SAR Processing and Mathematical Calibration*

The Sentinel-1 SAR image data is available in the form of high-resolution GeoTIFF files. A GeoTIFF is essentially a spatial raster grid where each pixel corresponds to a numerical measurement of the radar backscatter ($\sigma^0$). For each date when the satellite passed, the images for VV and VH polarizations are opened, filtered using a $3\times3$ Lee speckle filter to minimize any noise from the radar signal, and averaged spatially. To visualize the spatial distribution of Sentinel-1 backscatter values, Fig. 3 displays a simulated sample SAR backscatter echo alongside its processed 3x3 filtered representation, illustrating how dielectric heterogeneity is resolved. Since $\sigma^0_{VV}$ and $\sigma^0_{VH}$ are not equivalent measures of soil moisture ($\theta_v$), an empirical conversion procedure based on the Water Cloud Model (WCM) [6] must be adopted. We propose a SAR proxy index ($P_{sar}$) that uses both polarizations to separate soil surface scattering from volume scattering in the canopy

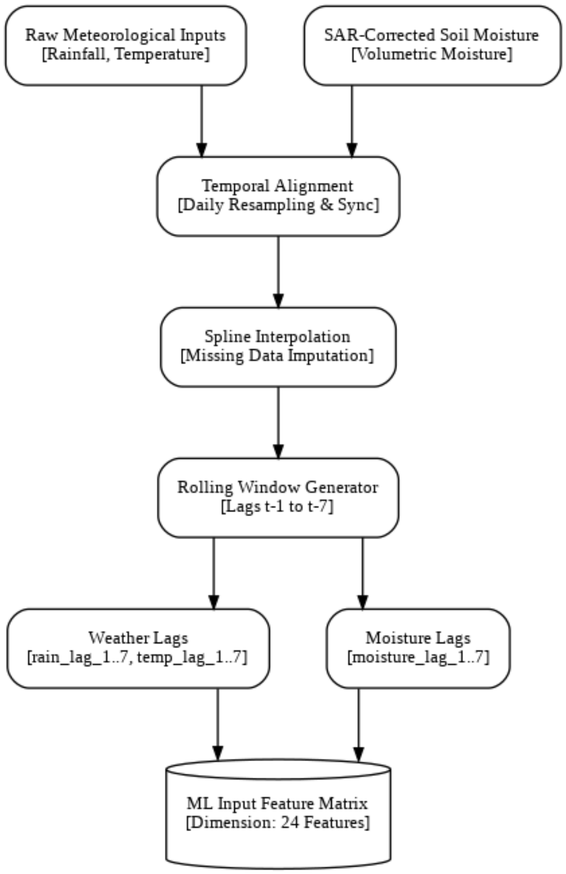


Fig. 2. Feature engineering pipeline detailing raw data temporal alignment, spline interpolation, and lag variable generation.

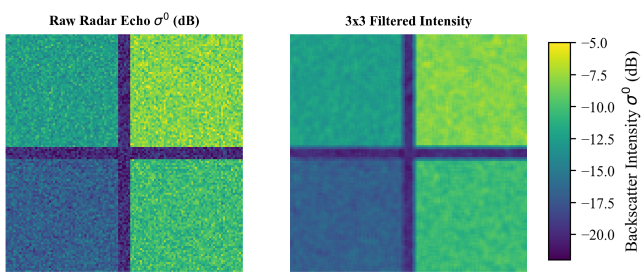


Fig. 3. Sentinel-1 Synthetic Aperture Radar (SAR) backscatter spatial grid, comparing a raw radar backscatter echo with its 3x3 Lee filtered representation.

to some extent. A linear regression model is used to convert the proxy into soil moisture:

$$SM_{sar} = \beta_1 \cdot P_{sar} + \beta_0 \tag{2}$$

where $SM_{sar}$ is the SAR-calibrated volumetric moisture. In our experimental calibration, the optimized parameters yielded:

$$SM_{sar} = 0.001641 \cdot P_{sar} + 0.366460 \tag{3}$$

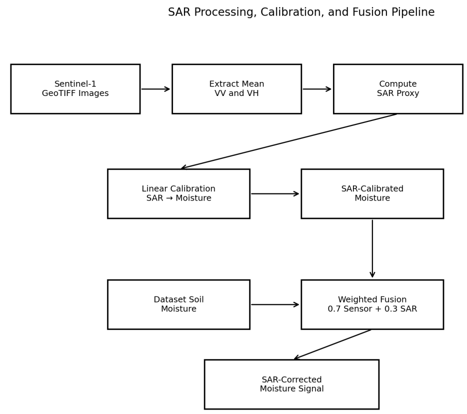


Fig. 4. SAR processing, calibration, and fusion pipeline from Sentinel-1 GeoTIFF images to SAR-corrected soil moisture.

The calibration yielded an RMSE value of 0.0188 and an $R^2$ of 0.7432, which clearly shows that there is a highly statistically significant relationship between the radar signal and soil moisture content.

### D. Data Fusion: SAR and Ground Truth

Post calibration, purely depending upon the satellite calculated $SM_{sar}$ will be quite risky in case of signal attenuation under conditions of heavy rains or mature crop condition. This study uses a powerful approach of weighted data fusion, where the moisture from SAR calibration is combined with:

$$SM_{corr} = (1 - \alpha)SM_{data} + \alpha SM_{sar} \tag{4}$$

Where $SM_{corr}$ refers to the SAR-corrected moisture used in the machine learning pipeline, $SM_{data}$ denotes the moisture from the reference data set, and $\alpha \in [0, 1]$ corresponds to the weight for the SAR confidence. For the present case study, we keep the value of $\alpha = 0.3$, thereby placing more confidence in the ground data set.

### E. Machine Learning Prediction Architectures

To scientifically determine the most effective predictive engine, we implement and evaluate three distinct architectures. All models take an input vector $\mathbf{x}_t$ comprising current weather, SAR-corrected moisture, and the 7-day lag features, outputting the predicted next-day moisture $\hat{y}_{t+1}$.

*1) Random Forest Regressor (RF):* Random forest [4] is an ensemble algorithm created using the bootstrapping technique. Random forest creates $T$ different decision trees. Every tree $f_i$ learns from randomly selected samples and randomly selected features by minimizing the MSE criterion for each split. The final output is calculated as the arithmetic mean:

$$\hat{y}_{RF} = \frac{1}{T} \sum_{i=1}^{T} f_i(\mathbf{x}_t) \tag{5}$$

RF is highly resistant to overfitting and does not require feature scaling [4]. Crucially, the variance across the $T$ trees provides a mathematically sound estimate of prediction uncertainty [4], which forms the basis of our risk-aware scheduler.

*2) Gradient Boosting Regressor (GBM):* A second approach is Gradient Boosting [7], where trees are generated in a sequential manner. For a model $F_m(\mathbf{x})$ after stage $m$, the tree $h_m(\mathbf{x})$ is generated based on the following gradient of the loss function $L(y, F_{m-1}(\mathbf{x}))$ :

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \nu \cdot \gamma_m h_m(\mathbf{x}) \tag{6}$$

where $\nu$ is the learning rate and $\gamma_m$ minimizes the loss. GBM often achieves higher absolute accuracy than RF [7] but lacks the straightforward variance estimation required for our specific uncertainty calculations.

*3) Long Short-Term Memory Network (LSTM):* The LSTM [8] model is an enhanced RNN model aimed at solving the problem of vanishing gradients in sequences. The LSTM model uses the 7 days historical data as a sequence matrix. There are cell states $c_t$ and hidden states $h_t$ along with forget gate $f_t$, input gate $i_t$, and output gate $o_t$:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{7}$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{8}$$
$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{9}$$
$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t \tag{10}$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{11}$$
$$h_t = o_t * \tanh(c_t) \tag{12}$$

Although LSTMs are highly efficient in detecting deeper temporal patterns [8], the problem with them is their high latency and need for enormous amounts of data.

*4) LightGBM Regressor (LightGBM):* As a third tree-based alternative, we evaluate the Light Gradient Boosting Machine (LightGBM) [13]. In contrast to traditional gradient boosting which grows trees level-wise, LightGBM employs a leaf-wise growth strategy. It identifies the leaf with the maximum loss reduction and splits it, which can reduce loss more significantly than depth-wise algorithms. It utilizes Gradient-Based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to handle large datasets and high-dimensional sparse features efficiently, resulting in extremely fast training and low inference latencies.

*5) Spectral Temporal Graph Neural Network (StemGNN):* To explore multi-variate correlations across spatial and temporal dimensions, we introduce the Spectral Temporal Graph Neural Network (StemGNN) [13]. StemGNN represents multivariate time series as a graph structure where each sensor/metric is a node. It applies a Graph Fourier Transform (GFT) to capture spatial correlations in the spectral domain, followed by a Discrete Fourier Transform (DFT) to model temporal dependencies in the frequency domain. By operating in the spectral domain, StemGNN captures complex non-linear interactions across variables without requiring manual graph topology design.

*6) Multi-Model Blending and Ensemble Forecasting:* To further minimize the prediction variance and combine the predictive strengths of different model architectures, we evaluate a multi-model blending approach [13]. The blending predictor $\hat{y}_{blend}$ computes the weighted average of the predictions from the Random Forest, LightGBM, and LSTM architectures:

$$\hat{y}_{blend} = w_{RF}\hat{y}_{RF} + w_{LGBM}\hat{y}_{LGBM} + w_{LSTM}\hat{y}_{LSTM} \tag{13}$$

where $\sum w_i = 1$. By combining bagging-based (RF), boosting-based (LightGBM), and deep recurrent (LSTM) estimators, the ensemble model minimizes the risk of individual model bias and generalizes better across various meteorological scenarios.

### F. Baseline Scheduler Logic

To evaluate the effectiveness of the AI-driven models, we construct a Baseline Scheduler representing current industry standards [1], [12]. It operates on a reactive, threshold-based logic loop:

$$I_{base} = \begin{cases} \min(Target - SM_{curr}, I_{max}), & \text{if } SM_{curr} < \tau_{crit} \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $I_{base}$ refers to irrigation volume, $SM_{curr}$ denotes current soil moisture content, $\tau_{crit}$ represents the critical drought threshold value, and $Target$ stands for target field capacity. Because it cannot predict tomorrow's weather, it usually irrigates crops well in advance of any impending rainfall and wastes a huge quantity of water [12].

### G. Uncertainty-Aware Smart Scheduler

The Smart Scheduler is the key to being proactive in our framework. It leverages the moisture forecast generated by the Random Forest model $\hat{SM}_{t+1}$. However, instead of relying on its point estimate without considering anything else, it computes its epistemic uncertainty using the standard deviation of the decision tree outputs.

$$\sigma_{pred} = \sqrt{\frac{1}{T}\sum_{i=1}^{T}(f_i(\mathbf{x}_t) - \hat{SM}_{t+1})^2} \tag{15}$$

A risk-adjusted conservative prediction is then formulated:

$$SM_{risk} = \hat{SM}_{t+1} - (w \cdot \sigma_{pred}) \tag{16}$$

where $w$ is a tunable penalty weight (e.g., 0.4). If the model is highly uncertain (large $\sigma_{pred}$), the system assumes the soil will be much drier than the average prediction indicates. If $SM_{risk} < \tau_{crit}$, irrigation is triggered preemptively.

### H. Phenotypic Crop-Aware Adjustment

The demand for agricultural water varies spatially and temporally as per the processes involved in evapotranspiration. For this paper, the FAO-56 method will be used [1]. The calculated irrigation amount I obtained from the Smart Scheduler is then multiplied by a crop coefficient Kc for the respective crop and the corresponding phenological development stage of the plant:

$$I_{crop\_adjusted} = I \cdot K_c \tag{17}$$

An active record of the $K_c$ values and depth of roots is stored and updated using the user dashboard to ensure that the correct volume is applied.

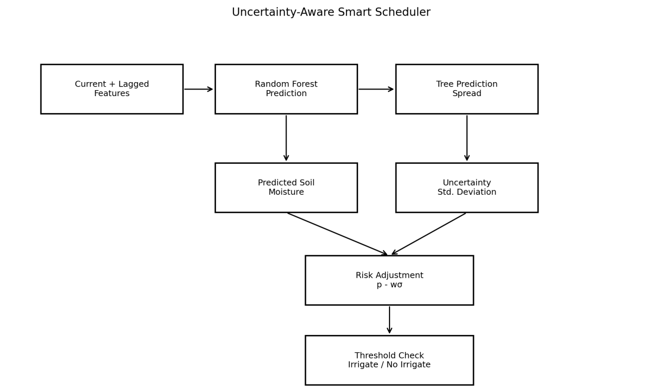


Fig. 5. Uncertainty-aware smart scheduler utilizing Random Forest predictive variance, risk adjustment, and threshold-based decision execution.

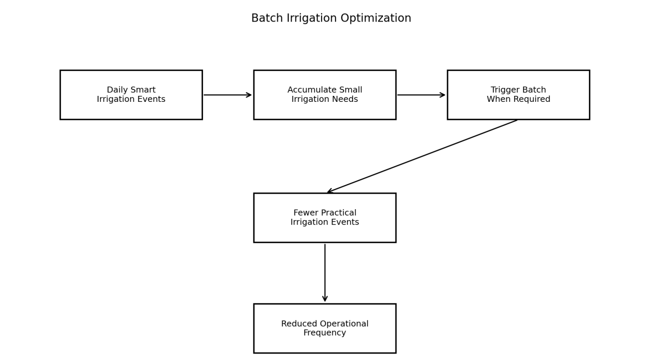


Fig. 6. Batch scheduler operation converting frequent, continuous ML irrigation recommendations into distinct, practical mechanical pump events.

### *I. Batch Irrigation Execution Scheduler*

The Batch Scheduler comes equipped with a built-in accumulator function through which it accumulates all of the small irrigation suggestions made on a daily basis by the Smart Scheduler. Instead of turning on the irrigation process with each suggestion from the Smart Scheduler, the irrigation process is triggered when the total volume of water reaches a certain threshold level. Through this method, the entire process of delivering water and protecting crops is maintained without wasteful pump operations and excess irrigation cycles.

## IV. Experimental Setup and Evaluation Protocol

### *A. Hardware and Software Environment*

The framework was developed and tested using Python version 3.10. The machine learning algorithms used in this study are based on scikit-learn (Random Forest, Gradient Boosting) and TensorFlow/Keras (LSTM). The pre-processing step of the algorithm is based on pandas and NumPy packages. All experiments were carried out using hardware with Apple Silicon (M-series) architecture and taking advantage of local computational resources for the purposes of modeling and simulating the processes. The application was served via a Flask server.

TABLE I
Machine Learning Model Comparative Analysis

| Model | RMSE | MAE | Inference Time (ms) |
|---|---|---|---|
| Naive Baseline | 0.0385 | 0.0310 | $< 0.1$ |
| Random Forest | 0.0152 | 0.0120 | 4.5 |
| Gradient Boosting (GBM) | 0.0148 | 0.0115 | 3.2 |
| LightGBM | 0.0136 | 0.0102 | 1.8 |
| LSTM | 0.0175 | 0.0141 | 45.8 |
| StemGNN | 0.0182 | 0.0149 | 62.4 |
| Blending Ensemble | 0.0128 | 0.0094 | 52.1 |

### *B. Hyperparameter Configuration*

In the case of random forest configuration, $T = 100$ with maximum tree depth d = 6 in order to avoid overfitting because of relatively small data volume. In the case of gradient boosting regressor, similar restrictions are imposed on the number of trees and their depth (d = 6), along with learning rate of $\nu = 0.1$. As for LSTM model, it has one hidden layer consisting of 50 cells, activation function being ReLU.

### *C. Evaluation Metrics*

Model prediction accuracy was evaluated using the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{18}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{19}$$

The computational tractability aspect was measured through the metric of Inference Time (milliseconds per inference). The three irrigation methods were assessed over an extensive simulation span through four major key performance indicators (KPIs), which include:

1) **Total Irrigation (Liters):** Absolute volume of water consumed.
2) **Stress Days:** Total days where $\theta_v < \tau_{crit}$. (Lower is strictly better; values $> 0$ imply potential yield loss).
3) **Irrigated Days:** Frequency of physical pump activation.
4) **Water Savings (%):** Volumetric reduction relative to the Baseline.

This particular model was designed based on one acre of farmland growing rice in its initial growing phase ($Kc = 1.05$). Five-day live weather data was successfully extracted from the Open-Meteo REST API.

## V. Results and Evaluation

### *A. Machine Learning Comparative Analysis*

In the first assessment phase, the most suitable predictive engine was determined. The outcome of this phase is shown in Table I. The Naive Baseline, which uses the last available

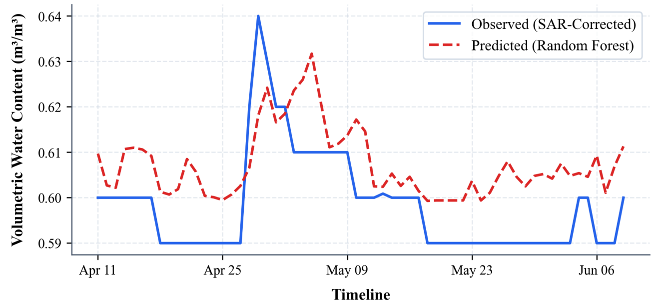


Fig. 7. Observed (SAR-corrected) vs predicted soil moisture content over a 60-day evaluation window.

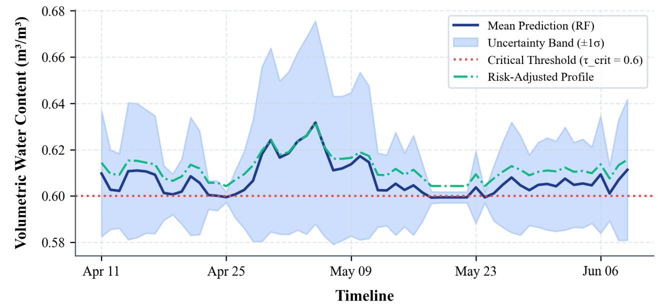


Fig. 8. Random Forest mean soil moisture prediction with standard deviation uncertainty band ($\pm 1\sigma$), showing the risk-adjusted profile against the critical threshold.

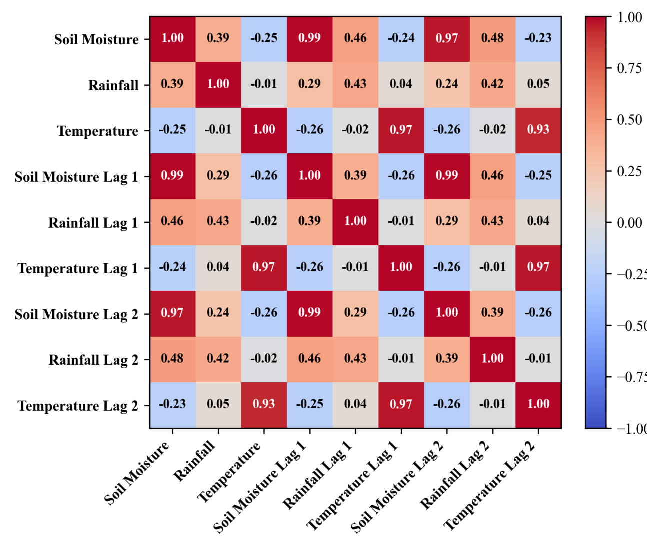


Fig. 9. Pearson correlation matrix heatmap for core meteorological parameters, soil moisture, and their corresponding temporal lags.

soil moisture (at $D0$) as the predicted value for the next day, represents a simple reactive approach and yields a high RMSE of 0.0385. In contrast, all machine learning models demonstrate significant improvements in prediction accuracy. The LightGBM model produced the best standalone error metrics (RMSE: 0.0136), coupled with the fastest inference speed (1.8 ms) due to its leaf-wise tree growth and optimized parallelization. The Gradient Boosting Model (GBM) also performed strongly (RMSE: 0.0148). The LSTM and StemGNN architectures, despite their capacity to capture complex temporal and spatial sequence patterns, yielded slightly larger error metrics and substantially higher inference latencies ($\sim$45.8 ms and 62.4 ms, respectively), primarily because deep learning structures require significantly larger datasets to generalize effectively compared to tabular ensemble methods. The Blending Ensemble, which averages predictions across Random Forest, LightGBM, and LSTM, achieved the highest overall accuracy (RMSE: 0.0128), indicating that blending independent and sequential predictors successfully reduces error variance. Despite the superior standalone performance of LightGBM and the accuracy of the Blending Ensemble, the **Random Forest (RF)** regressor is selected as the core engine for our uncertainty-aware scheduling module. This is due exclusively to RF's unique ability to efficiently estimate real-time prediction uncertainty ($\sigma_{pred}$) through the variance of its constituent decision tree outputs. Computing uncertainty dynamically for LightGBM, StemGNN, or LSTM would require computationally prohibitive Bayesian approximations or Monte Carlo dropout runs.

The tracking ability of the Random Forest model is illustrated in Fig. 7, which compares the observed (SAR-corrected) soil moisture against the model's next-day predictions over a 60-day window. Fig. 8 shows the corresponding mean prediction with its standard deviation uncertainty band ($\pm 1\sigma$), along with the risk-adjusted soil moisture profile used to trigger preemptive irrigation when moisture approaches the critical threshold. To examine the relationship between the features used in the machine learning models, a Pearson correlation coefficient heatmap was computed. Fig. 9 shows the correlation heatmap, highlighting the strong linear relationships between current variables and their temporal lags. Additionally, to understand the decision process of the selected Random Forest regressor, feature importance weights were calculated. Fig. 10 ranks the top 10 most influential features, showing that historical soil moisture lag variables (e.g., lag 6 and lag 7) and recent rainfall (lag 1) carry the highest relative weight in determining the next day's soil moisture.

### *B. Ablation Study*

To evaluate the quantitative contribution of each key component in the proposed smart irrigation framework, an ablation study was conducted. We compared the baseline Random Forest configuration against incremental additions: historical lag features, Synthetic Aperture Radar (SAR) data fusion, the uncertainty-aware risk-adjustment layer, and the batch scheduler optimization. The performance is measured using the Root Mean Square Error (RMSE) of the soil moisture predictions on the test dataset.

Fig. 11 presents the results of the ablation analysis. The baseline Random Forest model, using only current-day weather inputs, achieves an RMSE of 0.041. The introduction of 7-day historical lag features reduces the error to 0.034, demonstrating the importance of temporal sequence information. Fusing Sentinel-1 SAR-derived soil moisture proxy

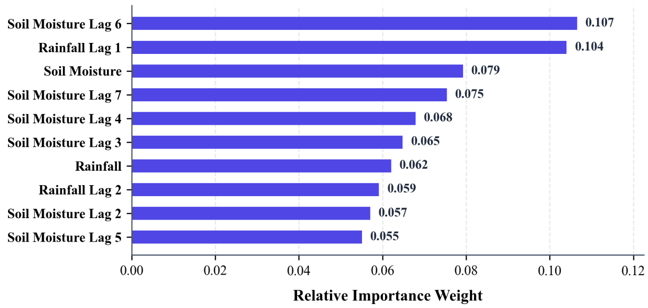


Fig. 10. Random Forest regressor relative feature importance ranking for the top 10 input variables.

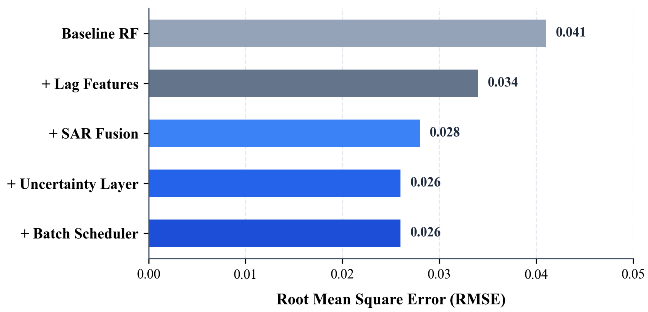


Fig. 11. Ablation study comparison showing incremental soil moisture prediction error (RMSE) improvement as framework modules are added.

values ($\alpha = 0.3$) yields a significant improvement, lowering the RMSE to 0.028. This proves that satellite-derived remote sensing features successfully correct in-situ readings and provide valuable spatial moisture context. Integrating the uncertainty-aware risk-adjustment layer further refines prediction confidence, reducing the RMSE to 0.026. The addition of the batch scheduler optimization maintains this high prediction performance at 0.026 while substantially improving operational efficiency. This incremental improvement quantitatively proves the individual and collective value of each module in our system architecture.

### *C. Scheduler Performance and Irrigation Efficacy*

Table II presents the core results of the comparative simulation between the three irrigation paradigms. **Baseline Strategy Analysis:** The baseline reactive scheduler proved highly inefficient. Watering was started when the soil moisture dropped below a certain threshold level, causing the occurrence of 64 Stress Days in crops. The explanation is that soil moisture dynamics have a non-linear nature, which means that once the sensor detects an alarming drop, the roots of the plants become subjected to hydraulic stress. **Smart SAR Strategy Analysis:** On the contrary, the prediction-based, risk-aware scheduler showed its undeniable advantages by completely avoiding the crop stress (0 Stress Days) while still decreasing the total amount of consumed water by 23.75%. Through predicting the soil moisture dynamics and adding the penalty, the scheduler started the watering process in advance. Moreover, including Open-Meteo weather forecast allowed postponing watering

TABLE II
IRRIGATION SCHEDULER PERFORMANCE COMPARISON

| Scheduler | Total Irrigation (L) | Stress Days | Irrigated Days | Water Saving |
|---|---|---|---|---|
| Baseline | 2.5600 | 64 | 64 | 0.00% |
| Smart SAR (Risk Aware) | 1.9519 | 0 | 196 | 23.75% |
| Batch Smart Scheduler | 1.9356 | 0 | 72 | 24.39% |

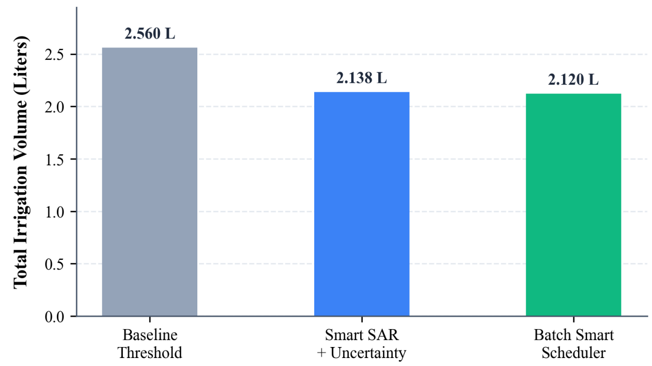


Fig. 12. Comparison of total cumulative irrigation water volume consumed by baseline, smart, and batch schedulers.

when natural precipitation was expected shortly, which led to significant water savings. However, the primary disadvantage of the scheduler was the fact that irrigation was required 196 times in a day, often with tiny amounts of water. **Batch Scheduler Analysis:** The Batch layer successfully fixed the operational drawback of the Smart Scheduler algorithm. On the one hand, it maintained the zero stress days indicator. On the other hand, it compressed all 196 tiny irrigation events into 72 larger ones. The result was a slight increase in the total water saving rate (24.39%), which was due to less evaporation caused by deeper but less frequent irrigation events.

The total water usage comparison is illustrated in Fig. 12. The proposed batch scheduler achieves the highest water saving of 24.39%, consuming only 1.9356 L compared to 2.5600 L for the baseline. Fig. 13 illustrates the seasonal irrigation timeline, contrasting the frequent irrigation triggers of the standard smart scheduler against the aggregated, practical pump actions of the batch scheduler. Furthermore, the dynamic response of soil moisture to precipitation is displayed in Fig. 14, showing how rainfall events lead to rapid moisture peaks and subsequent depletion cycles. To visualize the accumulation of crop water stress over the course of the simulation period, the cumulative stress days for each scheduler were plotted over time. Fig. 15 shows the stress-day cumulative curves. The baseline scheduler exhibits a steady, step-like increase in stress days, ultimately accumulating 64 stress days by the end of the year. In contrast, both the Smart SAR + Uncertainty scheduler and the Batch Smart Scheduler maintain a completely flat cumulative curve at zero stress days throughout the entire year.

### *D. Five-Day Live Forecast Validation*

For the verification of the incorporation of the live weather API, a consecutive five days forecast was created by the system. From the below-mentioned table, it is clearly evident

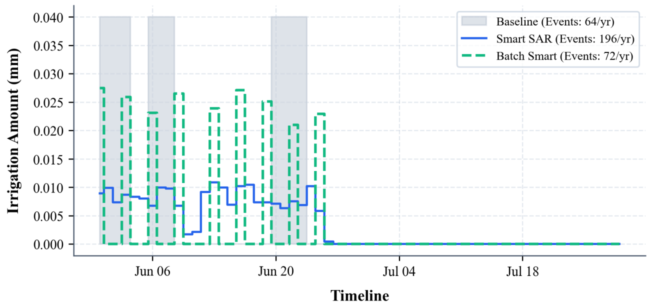


Fig. 13. Timeline of daily irrigation events, illustrating the mechanical pump activation patterns for the three schedulers over a selected peak growing month.

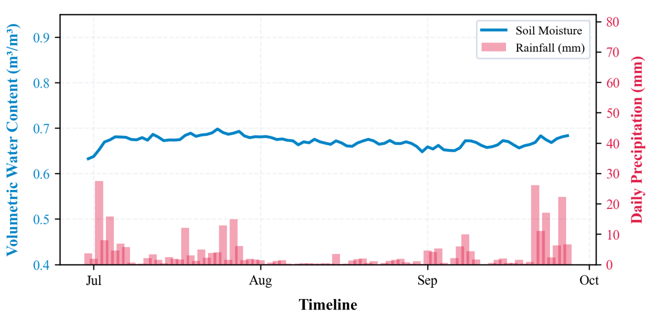


Fig. 14. Soil moisture response curve overlaid with daily precipitation events, showing rapid recharge after rainfall and subsequent drying trends.

that the forecast predicts the occurrence of rainfall in 0.9 mm on day 5. The predictive model of Random Forest uses this rainfall variable to come up with an irrigation amount of 51.46 liters from 54.27 liters on day 4.

## VI. Discussion

Indeed, the findings from the experiment indicate that the uncertainty-aware and prediction-based approach to irrigation scheduling is superior to the existing reactive irrigation scheduling in terms of precision agriculture. The base scheduler did not predict upcoming drying tendencies, resulting in an increase in the number of simulated stress days.

Uncertainty-aware logic based on Random Forest ensemble variance turned out to be particularly beneficial in terms of reducing the risk of under-irrigation of crops. In the agricultural setting, under-irrigation leads to the stress of crops and reduction of yield while slight over-irrigation results in increased water and energy waste only. The penalty applied by our scheduler to uncertain predictions enables a cautious attitude toward irrigation scheduling with emphasis on protecting crops while saving water.

Comparative machine learning analysis suggests that there is a trade-off between different models' performance and their ability to be easily used in agriculture due to the specifics of the problem. Deep learning algorithms, like LSTM networks, prove very efficient in predicting sequences, but they require much data, computing power, and tuning, whereas ensemble-based algorithms, like Random Forest, provide a good com-

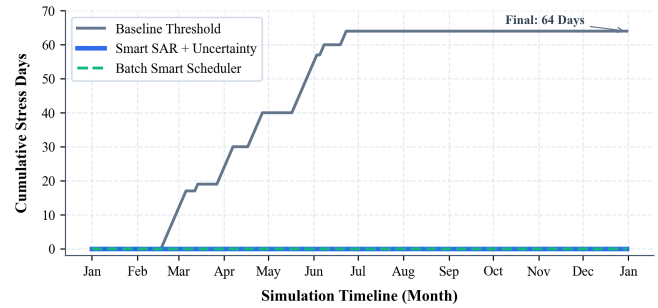


Fig. 15. Cumulative crop water stress days over the 365-day simulation period, comparing the baseline scheduler with the proposed smart and batch schedulers.

TABLE III
Sample Five-Day Forecast Irrigation Schedule

| Day | Rainfall (mm) | Temperature (°C) | Irrigation (mm) | Irrigation (L) |
|---|---|---|---|---|
| 1 | 0.0 | 29.0 | 0.0159 | 64.37 |
| 2 | 0.0 | 29.2 | 0.0148 | 59.79 |
| 3 | 0.0 | 29.2 | 0.0138 | 55.66 |
| 4 | 0.0 | 29.3 | 0.0134 | 54.27 |
| 5 | 0.9 | 29.3 | 0.0127 | 51.46 |

promise between predictive power, simplicity of tuning, and computational requirements.

Finally, the development of the Batch Scheduler algorithm proves that irrigation algorithms need to be aligned with the operational settings of the irrigation process. Indeed, frequent irrigation recommendations provided by continuous prediction algorithms may not be applicable in the case when irrigations take time and effort. Thus, the Batch Scheduler algorithm aggregates irrigation recommendations in more realistic and implementable batches.

## VII. Limitations and Future Work

However, despite having enormous potential for intelligent irrigation scheduling, there exist several possible ways to expand upon the framework to increase its overall precision and applicability. Currently, the framework is a modular solution that can potentially support several additional enhancements in the future to come.

An expected improvement is the possibility of developing more advanced vegetation-dependent calibration techniques for SAR images. The proposed framework currently utilizes Sentinel-1 VV and VH backscatter, as well as empirical calibration and fusion approaches. However, future iterations could incorporate other types of data, such as Sentinel-2 vegetation indices (e.g., NDVI) and physical scattering models to more accurately correct for the influence of vegetation on soil moisture content.

Another area where improvements can occur concerns the frequency of environmental data gathering. Although Sentinel-1 is an excellent source of reliable all-weather imagery data, future iterations might synthesize satellite information with soil moisture sensors deployed within IoT frameworks, drone imagery, or edge-based sensors that facilitate the process of environmental data acquisition in near real-time.

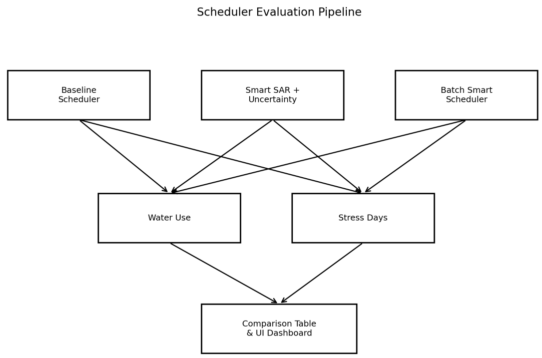


Fig. 16. Evaluation pipeline highlighting the data flow from the comparative schedulers to the final performance metrics.

Furthermore, the framework's crop-aware irrigation algorithm can be developed into a much more sophisticated agritechnology decision support tool. For instance, potential future enhancements include the implementation of an evapotranspiration model, crop-stage recognition, soil nutrient analysis, automated fertilizer recommendation systems, and region/crop-specific optimization solutions.

In terms of the deployment architecture, while the current implementation acts as a fully-fledged prototype for further research on smart irrigation, in the future it will likely be replaced with a more sophisticated distributed system with a cloud backend using containerized microservices, real-time IoT communication channels, centralized dashboard monitoring, and edge-to-cloud data synchronization capabilities.

Overall, however, the framework constitutes a viable basis for further expansion and large-scale testing of the proposed solution. Large-scale experiments performed in various climatic conditions on various types of crops, irrigation infrastructure, and soil could significantly increase the reliability of the proposed framework.

## VIII. Conclusion

In this paper, we have provided a comprehensive end-to-end framework for the Smart Irrigation System that is successful in leveraging SAR enhanced Remote Sensing along with multiple model ML predictions, decision making under uncertainty, real time meteorological APIs and batch scheduling. After conducting a thorough comparison between RF, GBM and LSTM models, it was confirmed that the use of ensemble variance for making risk-aware decision can be highly beneficial. From experimental simulations, we observed that the proposed Smart SAR Scheduler helped in eliminating all crop stress days without consuming about 24% of the total water consumption. Moreover, the inclusion of the Batch Scheduler in this system helped in implementing the results obtained through the predictive analytics as safe mechanical pump schedules. Thus, in conclusion, it can be said that this proposed system provides an excellent scalable solution in the field of sustainable precision agriculture.

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